

INDUSTRIAL INTERNSHIP REPORT

on

“Student Performance Analysis”

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The IoT Academy
in collaboration with UniConverge Technologies Pvt Ltd (UCT)

Executive Summary

This report presents the work completed during the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). The internship provided an opportunity to understand practical software development and to work on a project based on Python programming and data analysis.

My project, “Student Performance Analysis”, is a console-based Python application developed using Pandas and NumPy. The system accepts student details, subject marks and attendance from the user and processes the data to generate total marks, percentage, average, result status, highest scorer, class average and average attendance. Pandas is used for structured tabular data processing through DataFrames, while NumPy is used for numerical calculations.

The internship helped me connect programming concepts with a practical problem, improve my coding skills, understand data processing, test a working program and document the complete project. It also provided exposure to an industry-oriented workflow and improved my confidence in Python programming.

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1. Preface

The six-week Industrial Internship was an important opportunity to gain practical exposure to software development. Academic learning provides programming fundamentals, while an internship helps apply those concepts to a practical problem. During this internship, I worked on a Student Performance Analysis project using Python, Pandas and NumPy.

The project was designed to accept student information and marks, organize the data into a structured DataFrame and calculate useful academic statistics. The work involved understanding the problem, writing the Python program, using external libraries, testing different inputs and preparing the project documentation.

The program was planned in stages: understanding requirements, collecting inputs, creating the DataFrame, performing calculations, generating the result and testing the output. The internship improved my programming, data-handling, debugging, documentation and problem-solving skills.

I sincerely thank upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd, my internship mentors and all faculty members who supported and guided me during this internship. I also thank my peers and juniors for their encouragement. I recommend that students use internship opportunities to practice concepts on real or realistic problems and maintain proper documentation throughout the project.

2. Introduction

Student performance analysis is the process of organizing and interpreting academic information such as marks, percentages, results and attendance. Manual calculations for several students can become repetitive and may lead to calculation errors. A simple Python-based system can automate these calculations and present the information in a structured format.

The developed application is a command-line project. The user enters the number of students and then enters each student's name, four subject marks and attendance percentage. The application stores the data in a Pandas DataFrame and automatically calculates total marks, percentage, average, pass/fail result, highest scorer, class average and average attendance.

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is described in the internship material as a company established in 2013 and working in the Digital Transformation domain, with a focus on sustainability and return on investment. Its technology areas include Internet of Things (IoT), Cyber Security, Cloud Computing, Machine Learning, communication technologies such as 4G/5G/LoraWAN, Java Full Stack, Python and front-end technologies.

UCT's technology and solution areas include IoT platforms, smart factory solutions, LoRaWAN-based applications and predictive maintenance. These areas demonstrate the importance of software, data processing and automation in industrial applications.

2.2 About upskill Campus (USC)

upskill Campus, together with The IoT Academy and in association with UniConverge Technologies, facilitated the internship process. The program provided a structured environment for students to learn technologies, work on a project and document their learning.

2.3 The IoT Academy

The IoT Academy is the EdTech division associated with UCT and runs executive certification and learning programs in multiple technology domains. In the internship context, it supported the learning and execution process.

2.4 Objectives of the Internship

- To gain practical experience in software development.
- To apply Python programming concepts to a practical problem.
- To learn data handling and analysis using Pandas and NumPy.
- To develop problem-solving and debugging skills.
- To understand basic project testing and documentation.
- To improve communication, technical writing and confidence.

2.5 Reference

1. Upskill Campus internship instructions and project material.
2. Python documentation and learning material.
3. Pandas documentation and learning resources.
4. NumPy documentation and learning resources.

2.6 Glossary

Term	Meaning
Python	High-level programming language used to develop the application.
Pandas	Python library used for data manipulation and tabular analysis.
NumPy	Python library used for numerical computation.
DataFrame	Two-dimensional labeled data structure provided by Pandas.
Percentage	Marks expressed out of the maximum marks as a percentage.
Attendance	Student attendance expressed as a percentage.
CSV/Data Table	Structured representation of rows and columns of data.

3. Problem Statement

The problem is to develop a simple Student Performance Analysis system that can accept academic data for multiple students and automatically calculate important performance measures. The system should reduce repetitive manual calculations and present the final information in a structured table.

The required inputs are student name, marks in Mathematics, Science, English and Computer, and attendance percentage. From these inputs, the system should calculate total marks, percentage, average, result status, highest scorer, class average and average attendance.

4. Existing and Proposed Solution

Existing Solution

In a basic manual approach, teachers or students may calculate totals, percentages and averages using paper, a calculator or separate spreadsheet formulas. While this can work for a small number of students, repeated calculations can take time and may introduce human calculation errors.

Proposed Solution

The proposed solution is a Python console application using Pandas and NumPy. It accepts data through the keyboard, stores the information in a DataFrame and performs calculations programmatically. The final DataFrame displays the complete result in one table.

The proposed solution adds value through automation, consistent calculations, reusable code and easy extension. Additional features such as CSV export, graphical reports, database storage or a graphical user interface can be added in future.

4.1 Code Submission (GitHub Link)

[upskillcampus/studentPerformanceAnalysis.py at main · mayurimore5222-netzin/upskillcampus](https://github.com/mayurimore5222-netzin/upskillcampus-studentPerformanceAnalysis.py)

4.2 Report Submission (GitHub Link)

5. Proposed Design / Model

The system follows a simple input-process-output model. First, the user enters the number of students and their details. Next, the application stores the data and performs calculations. Finally, the application displays the result table and summary statistics.

5.1 High Level Flow

Start → Enter number of students → Enter student details → Create DataFrame → Calculate Total → Calculate Percentage → Calculate Average → Determine Pass/Fail → Find Highest Scorer → Calculate Class Average → Calculate Average Attendance → Display Results → End

5.2 Low Level Flow

1. Import Pandas and NumPy.
2. Read the number of students.
3. Repeat input collection for each student.
4. Store each student's data in a list.
5. Convert the list into a Pandas DataFrame.
6. Sum the four subject marks to calculate Total.

7. Divide Total by 400 and multiply by 100 to calculate Percentage.
8. Calculate Average using NumPy.
9. Assign Pass when Percentage is at least 40; otherwise assign Fail.
10. Identify the row with the maximum Percentage.
11. Calculate class average and average attendance.
12. Display the complete result.

5.3 Interfaces

The current project uses a command-line interface (CLI). The user interacts with the application through text prompts such as entering the student name, subject marks and attendance. The final output is displayed as a formatted table.

Example interface sections include: application heading, student input prompts, Student Result table, Highest Scorer section, Class Average and Average Attendance.

6. Performance Test

Performance testing for this project focuses on correctness, input processing and calculation consistency. Since the application is a lightweight console program, its main practical constraints are valid numeric input, correct maximum marks and the number of students being processed.

6.1 Test Plan / Test Cases

Test Case	Input / Condition	Expected Result	Status
TC01	One student with valid marks	Correct total, percentage and average	Pass
TC02	Multiple students	All student records displayed	Pass
TC03	Percentage ≥ 40	Result displayed as Pass	Pass
TC04	Percentage < 40	Result displayed as Fail	Pass
TC05	Different percentages	Highest scorer identified correctly	Pass
TC06	Different attendance values	Average attendance calculated	Pass

6.2 Test Procedure

5. Run the Python program in a Python-supported environment.
6. Enter the number of students.
7. Enter valid names, marks and attendance values.
8. Check the generated DataFrame.
9. Manually verify selected totals and percentages.
10. Verify the Pass/Fail result for boundary and low-score cases.
11. Compare the highest scorer with the entered percentages.
12. Verify class average and average attendance.

6.3 Performance Outcome

The program successfully performs the intended calculations for the supplied student records. Pandas provides a convenient tabular structure for displaying and processing multiple student records, while NumPy provides numerical operations for averages. The design is lightweight and suitable for a small academic dataset. For larger datasets, validation, file/database storage and optimized input handling would be useful improvements.

7. My Learnings

- Improved understanding of Python programming and program structure.
- Learned how to use Pandas DataFrames for tabular data.
- Learned how NumPy can be used for numerical calculations.
- Understood how to collect repeated user input using loops.
- Learned to calculate and interpret totals, percentages and averages programmatically.
- Improved debugging and testing skills.
- Understood the importance of project documentation and GitHub submission.
- Improved confidence in explaining a project and its workflow.

Overall, the internship connected theoretical programming knowledge with a practical project and helped me understand how a small software solution can be planned, implemented, tested and documented.

8. Future Work Scope

- Add input validation so marks remain within an appropriate range.
- Allow import of student data from CSV or Excel files.
- Export the generated result to CSV/PDF.
- Add charts for subject-wise and student-wise performance.
- Add a graphical user interface using Tkinter or a web interface.
- Store student records in a database.
- Add login and role-based access for teachers and administrators.
- Generate detailed student report cards automatically.

Appendix: Project Technology

- Programming Language: Python
- Libraries: Pandas, NumPy
- Interface: Command Line Interface (CLI)
- Data Structure: Pandas DataFrame
- Project Type: Student Performance Analysis

Appendix: GitHub Submission Checklist

13. Create GitHub repository named upskillcampus.
14. Upload StudentPerformanceAnalysis.py.
15. Create and finalize the internship report PDF.
16. Add the GitHub code link inside the report.
17. Upload the final report PDF to the same repository.
18. Copy the final working GitHub links.
19. Submit the required link/details on the Upskill Campus portal.